Privacy-Enhanced Safe Reinforcement Learning for the Dispatch of a Local Energy Community

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***Abstract*—Local Energy Community (LEC) has emerged as a viable community-focus framework to enhance local reliability and energy efficiency by integrating different energy sectors and managing local distributed energy resources (DERs). However, the difficulties associated with handling model complexity, along with privacy concerns arising from interactions between energy operators within the LEC, pose challenges for traditional algorithms in achieving coordinated dispatch. To this end, we develop a novel privacy-enhanced, safe, coordinated dispatch framework that integrates reinforcement learning (RL), the perturbation module, and the safety module. The private states of each energy sector within the LEC are concealed by the independent perturbation module before sharing. A central RL agent is then trained on the concealed state space to learn the optimal policy for coordinated dispatch under the complex and uncertain environment. Furthermore, dispatch actions are evaluated and refined by the safety module before the operators execute them. In this way, we can obtain an optimal policy without disclosing any sector's private state while ensuring the safe operation of the LEC. Extensive experiments are carried out to validate the superior performance and scalability of the proposed method.**

***Index Terms*—Local energy community, coordinated dispatch, privacy enhancement, safe operation, reinforcement learning.**

NOMENCLATURE

Sets

T	Set of time steps
$\mathcal{G}, \mathcal{T}$	Set of diesel generators (DGs), gas turbines (GTs)
$\mathcal{M}, \mathcal{B}$	Set of electric heat pumps (EHPs), gas boilers (GBs)
$\mathcal{W}$	Set of gas wells (GWs)
$\mathcal{C}^p, \mathcal{C}^g, \mathcal{C}^h$	Set of energy storages (ESs), gas storages (GSs), thermal energy storages (TSs)
$\mathcal{L}^p, \mathcal{L}^g$	Set of power pipelines (PPs) and gas pipelines (GPs)
$\mathcal{L}^s, \mathcal{L}^r$	Set of supply water pipeline and return water pipeline
$\mathcal{N}^p, \mathcal{N}^g, \mathcal{N}^h$	Set of power buses (PBs), gas nodes (GNs) and heat nodes (HNs)
$\mathcal{N}^s, \mathcal{N}^l$	Set of HNs with heat source/load

Parameters

$C_t^{\mathrm{PS}}, C_t^{\mathrm{GS}}, C_t^{\mathrm{HS}}$	Operational costs for power system/gas system/heat system
$\lambda_{\mathrm{buy},t}^{p/q}, \lambda_{\mathrm{sell},t}^{p/q}$	Price of buying/selling active/reactive power with the main grid
$c_{g,t}^{p}, c_{g,t}^{q}$	Generation cost of active/reactive power
λ_t^{gas}	Trading cost of natural gas from the main gas grid
$c_{g,t}^{\mathrm{gas}}$	Cost of gas generation of generator g
$c_{g,t}^{\mathrm{heat}}$	Cost of heat generation of generator g
$\overline{P}_g, \underline{P}_g$	Upper and lower bound for active power of unit g
$\overline{Q}_g, \underline{Q}_g$	Upper and lower bound for reactive power of unit g
σ_g	Rated power factor of unit g
$\eta_g^{\mathrm{gt}}, \eta_g^{\mathrm{ehp}}, \eta_g^{\mathrm{gb}}$	Energy conversion efficiency of GT/EHP/GB g
$\overline{S}_m^{\mathrm{es}}, \overline{S}_m^{\mathrm{gs}}, \overline{S}_m^{\mathrm{ts}}$	Maximum SoC of ES/GS/TS m
$\underline{S}_m^{\mathrm{es}}, \underline{S}_m^{\mathrm{gs}}, \underline{S}_m^{\mathrm{ts}}$	Minimum SoC of ES/GS/TS m
$E_m^{\mathrm{es}}, E_m^{\mathrm{gs}}, E_m^{\mathrm{ts}}$	Capacity of ES/GS/TS m
$\eta_m^{\mathrm{esc}}, \eta_m^{\mathrm{gsc}}, \eta_m^{\mathrm{tsc}}$	Charging efficiency of ES/GS/TS m
$\eta_m^{\mathrm{esd}}, \eta_m^{\mathrm{gsd}}, \eta_m^{\mathrm{tsd}}$	Discharging efficiency of ES/GS/TS m
$\overline{P}_m^{\mathrm{es}}, \overline{F}_m^{\mathrm{gs}}, \overline{H}_m^{\mathrm{ts}}$	Charging capacity of ES/GS/TS m
r_{ij}, x_{ij}	Resistance and reactance of PP ij
$\overline{S}_{ij,t}$	Capacity of PP ij
$p_{j,t}^D, q_{j,t}^D$	Active/reactive power demand of PB j
U_0	Voltage magnitude in the slack bus (1 p.u.)
$\overline{V}, \underline{V}$	Upper and lower bound for nodal voltage
$\overline{F}_g^{\mathrm{gw}}, \underline{F}_g^{\mathrm{gw}}$	Upper and lower bound for gas output of GW g
$f_{j,t}^D, H_{j,t}^D$	Gas/heat demand of GN/HN j
$\overline{\pi}_{b,t}, \underline{\pi}_{b,t}$	Upper and lower bound for nodal pressure of GN b
$\overline{F}_{bp}, \underline{F}_{bp}$	Upper and lower bound for gas flow of GP bp

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α_{bp}	Compression factor of the compressor of GP bp.
β_{bp}	Parameters for gas flow and pressure relationship of GP bp
$\overline{H}_g$	Maximum heat output of heat generator g
T^{amb}	Ambient temperature
κ_{ij}, d_{ij}	Heat transfer coefficient and length of the water pipeline ij
c	Specific heat capacity
m_{ij}	Mass flow of water pipeline ij
$\underline{T_j^s}$, $\underline{T_j^r}$, $\overline{T_j^s}$, $\overline{T_j^r}$	Limit of supply/return temperature

Variables

$P^{PG}_{\text{buy},t}$, $P^{PG}_{\text{sell},t}$	Active power exchanged with the main grid
$Q^{PG}_{\text{buy},t}$, $Q^{PG}_{\text{sell},t}$	Reactive power exchanged with the main grid
$P_{g,t}$, $Q_{g,t}$	Active/reactive power generated by generator g
F_t^{GG}	Gas supply from the main gas grid
$F_{g,t}$	Gas output of GW g
$H_{g,t}$	Heat output of heat generator g
$F^{\text{gt}}_{g,t}$, $F^{\text{gb}}_{g,t}$	Gas consumption of GT/GB g
$S^{\text{es}}_{m,t}$, $S^{\text{gs}}_{m,t}$, $S^{\text{ts}}_{m,t}$	State of charge (SoC) of ES/GS/TS m
$\mathcal{I}^{\text{es}}_{m,t}$, $\mathcal{I}^{\text{gs}}_{m,t}$, $\mathcal{I}^{\text{ts}}_{m,t}$	Charging status of ES/GS/TS m
$P^{\text{ch}}_{m,t}$, $P^{\text{dis}}_{m,t}$	Charging and discharging power of ES m
$P_{ij,t}$, $Q_{ij,t}$	Active and reactive power flow of PP ij
$p^G_{j,t}$, $q^G_{j,t}$	Active/reactive power generation of PB j
$V_{j,t}$	Nodal voltage of PB j
$F^{\text{ch}}_{m,t}$, $F^{\text{dis}}_{m,t}$	Inflating and deflating gas of GS m
$\pi_{b,t}$	Nodal pressure of gas node b
$f^G_{j,t}$	Gas generation of GN j
$F_{bp,t}$	Gas flow of GP bp
$P^{\text{ehp}}_{g,t}$	Energy consumption of EHP g
$H^{\text{ch}}_{m,t}$, $H^{\text{dis}}_{m,t}$	Charging and discharging heat of TS g
$T^{\text{start}}_{ij,t}$, $T^{\text{end}}_{ij,t}$	Inflow and outflow temperature of water pipeline ij
$T^{\text{mix}}_{j,t}$	Mixed outflow temperature of HN j
$T^s_{j,t}$, $T^r_{j,t}$	Supply/return temperature of HN j
$H^G_{j,t}$	Heat generation of HN j

I. Introduction

THE GROWING integration of distributed energy resources (DERs) has prompted a shift in the demand side from traditionally passive to active systems [1], which poses significant challenges on local control and management solutions in terms of reliability and energy efficiency [2]. To tackle these issues, the local energy community (LEC) has emerged as a feasible framework for coordinating DERs within a specific community [3]. However, the challenges in handling model complexity, coupled with privacy concerns within the LEC, make the realization of coordinated dispatch intractable for conventional control algorithms.

The endogenous uncertainties of renewables and DERs within the LEC system greatly contribute to its model complexity. Substantial research endeavors have been dedicated to achieving coordinated dispatch for the LEC in this context. One commonly used approach is stochastic programming (SP) [4], which handles uncertainties by scenario-based methods. By taking into account the stochastic nature of the supply and demand sides, the efficient operation of the multi-energy system (MES) was achieved in [5] and [6] using SP and robust optimization approaches, respectively. In [7], the optimal scheduling problem in a residential microgrid, considering demand response, was addressed by a stochastic multi-objective optimization algorithm that aims to minimize both operation costs and emissions. In addition, model predictive control (MPC) [8] has been proven to be an effective alternative for coordinated dispatch. Reference [9] addressed the operation control of multi-energy microgrids by employing day-ahead SP and real-time sliding window MPC. Reference [10] proposed a bi-level MPC-based method combining rolling prediction and feedback correction to optimize the system scheduling. However, the above research may face challenges concerning scalability and real-time decision-making capacity as the complexity and size of the model expand. SP-based and MPC-based methods are brittle with respect to the accuracy of distribution modeling of uncertainties. As the number of scenarios considered for better capturing the uncertainty distribution increases, the complexity of the control problem escalates, thereby resulting in a substantial computational burden and rendering the problem-solving process considerably time-consuming. Although the scenario-reduction method [11] is regarded as a potential solution, it is still a challenging task to acquire an appropriate set of representative scenarios for identifying uncertainty distribution [12].

Reinforcement Learning (RL) has been regarded as an effective alternative to the aforementioned model-based methods in the realm of optimal control for MES. In uncertain environments, RL algorithms can dynamically adapt to changes and learn to make optimal decisions by continuously updating their policies based on observations. This allows RL agents to develop robust decision-making strategies without the need for explicit models or prior knowledge about the environment's dynamics. Once trained, RL algorithms can make decisions in real time, which is suitable for applications in dynamic and complex environments where rapid decision-making is crucial for success.

Both Single-Agent Reinforcement Learning (SARL) and Multi-Agent Reinforcement Learning (MARL) have been employed and achieved reasonable performance in the problem of coordinated dispatch for MES. Particularly when the number of stakeholders or participants involved in a cooperative problem is huge, MARL can serve as an efficient solution, allowing multiple agents to simultaneously learn and make decisions. This parallel nature of learning can greatly reduce computational complexity. In [13], a MARL algorithm on a device basis was proposed to efficiently manage energy in multi-energy industrial parks. Reference [14] employed a prosumer-based MARL algorithm to tackle the energy management problem within an energy community considering peer-to-peer trading among prosumers. Reference [15] proposed an adaptive MARL algorithm to effectively manage

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the multi-energy buildings within a virtual power plant (VPP) while considering the dynamic behaviors of agents joining or leaving the VPP. However, there are multiple significant concerns regarding the MARL application. Firstly, existing MARL algorithms, such as those in [13], [14], [15], typically use a Centralised Training with Decentralised Execution (CTDE) framework to manage non-stationarity, but this approach can lead to privacy issues as it requires access to global state and teammates' actions, potentially disclosing private information of each agent during training. Secondly, parameter sharing is commonly employed in many advanced MARL algorithms, as demonstrated in [14] and [15], which require agents to be homogeneous (i.e., share the same observation and action space). However, this technique is not consistent with the context of our work involving heterogeneous operators. Lastly, existing MARL algorithms implement simultaneous updates for agents. Under this update scheme, agents lack awareness of other agents' update directions, which could potentially provoke conflicting updates, leading to training instability and the failure of convergence [16]. To this end, we take a step back to SARL algorithms, which are also proven to be effective in handling multi-energy coordination problems [17], [18], [19]. But they also have issues regarding privacy and safe operation. In the following content, we will further discuss and review current progress regarding the safety and privacy issues of RL applications in the context of energy management for MES.

Conventional RL cannot guarantee the satisfaction of system constraints, primarily due to its unconstrained learning nature. As RL algorithms focus on maximizing cumulative rewards during training, they may inadvertently explore actions that violate the inherent constraints of the system. To this end, safe operation is one critical challenge for RL implementation. In [17], the authors proposed a penalty-based RL algorithm, which adds a penalty term for operational constraint violation into the reward function, aiming to achieve real-time scheduling of an MES coupled with carbon-capturing technology. In [18], a constrained RL algorithm based on a constrained Markov decision process (CMDP) model was proposed for optimal control of an integrated electricity and natural gas system (IEGS) considering the physical constraints. However, both the penalty-based and CMDP-based RL methods have a brittle performance with respect to the Lagrange multipliers and have no guarantee of satisfying the physical constraints during training. Although a neural network-based security layer enhanced operational safety during the RL training process in [19], issues regarding explainability and provable safety guarantees, stemming from the black-box model, hinder practical applications of the approach in the real world.

The realization of coordinated dispatch for LEC may involve multiple stakeholders. To obtain an optimal solution, the conventional RL paradigm necessitates the aggregation and concatenation of states containing confidential information from different energy sectors in a centralized manner. Consequently, individual operators might be reluctant to disclose their privacy, posing a challenge to effective collaboration and information sharing in this context.

To encourage operators' participation in coordinated dispatch, it is essential to incorporate a privacy-preserving mechanism for the states of operators during the training process. A comprehensive analysis of privacy-preserving techniques for RL was presented in [20], which highlighted several notable methods, including differential privacy (DP) [21] and cryptography [22]. The core idea behind DP is to add carefully calibrated noise to the data, thereby safeguarding sensitive information while still facilitating the extraction of meaningful insights from the data. However, the noise introduced by the DP mechanism can compromise the accuracy of state identification, subsequently degrading the performance of the RL algorithm. Cryptography also serves as the traditional technique employed in privacy protection domains, wherein messages are encoded to prevent unauthorized parties from deciphering their content. Homomorphic encryption (HE) [23] is among the most popular cryptographic techniques, allowing computations to be executed directly on encrypted data without necessitating decryption. Nevertheless, due to the inherent computational complexity and inefficiency of HE, performing operations on encrypted data demands considerably more time and computational resources, rendering the integration between RL and HE infeasible. To this end, identifying an appropriate privacy-preserving mechanism for protecting the private states that seamlessly aligns with the RL paradigm's training process is of paramount importance.

Motivated by the goal of efficiently addressing model complexity and handling safe operation and privacy issues, this paper proposed a privacy-enhanced safe RL framework to solve the coordinated dispatch problem for the LEC, which ensures the LEC's operational safety while simultaneously preserving the privacy of the operators.

The contributions of this paper are threefold:

- Propose a privacy-enhanced safe soft actor-critic (PSSAC) framework to tackle the novel problem of deploying RL algorithms in a manner that ensures both privacy and safety. In this framework, two types of innovative modules, namely safety modules and perturbation modules, empower the RL algorithm to achieve secure and privacy-enhanced LEC scheduling. The perturbation modules independently concealed the private states of operators before being utilized in RL agent's training process, thereby effectively preventing privacy leakage. Additionally, the safety modules, situated on the operator's side, safeguard the operation process and guide the learning process of the RL agent.
- Investigate a lightweight privacy-enhanced mechanism tailored for RL, which can be seamlessly integrated into the RL training process without imposing additional computational burdens or degrading the algorithm's performance. The perturbation process entails only straightforward matrix multiplication operations but preserves the characteristics of the state space (i.e., isometric transformation), which renders the mechanism both lightweight and well-suited for RL training. Moreover, a comprehensive security analysis of the proposed method is provided by analyzing a set of potential attacks. To the best of our knowledge, this is the first work to investigate the isometric perturbation mechanism for SARL.

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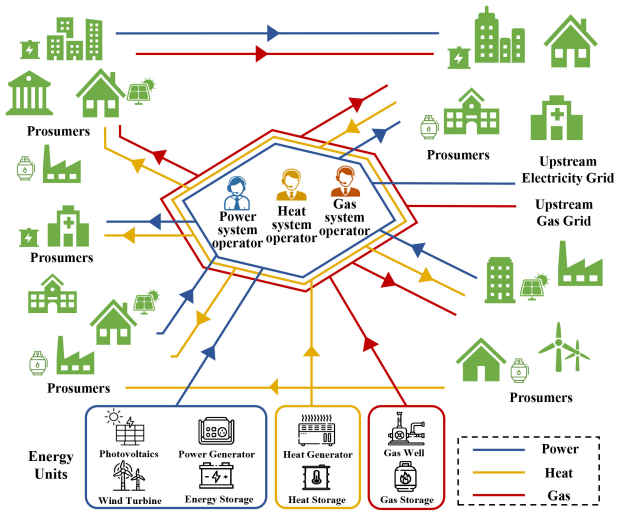


Fig. 1. Schematic of an LEC.

- Conduct comprehensive numerical experiments to prove that the model privacy and safe operation can be simultaneously maintained by adopting distributed safety modules to externalized operational constraints as part of the environment, which effectively guide the learning process of the RL agent and rectify infeasible actions. Furthermore, the privacy of operators' states can be adequately safeguarded by executing policy improvement within an isometrically transformed state space where the private information of each operator is blurred. The experimental result validates that the proposed method outperforms traditional algorithms and shows scalability on a large-scale LEC.

The rest of this paper is organized as follows: Section II introduces the problem to be solved and the corresponding formulated Markov decision process (MDP). Section III presents the proposed PSSAC framework. Section IV conducts case studies, while Section V draws the conclusion.

II. Problem Formulation

As illustrated in Fig. 1, an LEC contains a cluster of prosumers (buildings, factories, etc.) and a diverse range of energy units owned by different operators, including photovoltaic (PV) panels, wind turbines (WTs), energy generators, and storage systems.

The LEC mainly involves three types of system operators: the power system operator (PSO), gas system operator (GSO), and heat system operator (HSO). Each operator manages its own energy units and the corresponding network, which are detailedly modeled in II-A. We aim to realize coordinated dispatch while preserving privacy and ensuring the safety of three operators. Therefore, distributed state perturbation and action correction are performed before and after decision-making, respectively, which will be detailed in Section III.

In this section, an LEC dispatch model is established by fully considering the energy coupling relationship and network constraints. Moreover, the proposed coordinated problem is transformed into an MDP for the subsequent application of the RL paradigm.

A. LEC Dispatch Model

Each energy sector, along with its associated community-shared energy components, is managed by the respective system operator. The objective of coordinated LEC dispatch is to minimize the expectation of overall operational cost.

$$\min \sum_{t\in T}\mathbb{E}\left\{C_t^{\mathrm{PS}}+C_t^{\mathrm{GS}}+C_t^{\mathrm{HS}}\right\} \tag{1a}$$

$$\begin{aligned} C_t^{\mathrm{PS}} &= \lambda_{\mathrm{buy},t}^{p}P_{\mathrm{buy},t}^{\mathrm{PG}}+\lambda_{\mathrm{buy},t}^{q}Q_{\mathrm{buy},t}^{\mathrm{PG}} \\ &\quad+\lambda_{\mathrm{sell},t}^{p}P_{\mathrm{sell},t}^{\mathrm{PG}}+\lambda_{\mathrm{sell},t}^{q}Q_{\mathrm{sell},t}^{\mathrm{PG}} \\ &\quad+\sum_{g\in\mathcal{G}\cup\mathcal{T}}c_{g,t}^{p}P_{g,t}+\sum_{g\in\mathcal{G}\cup\mathcal{T}}c_{g,t}^{q}Q_{g,t} \end{aligned} \tag{1b}$$

$$C_t^{\mathrm{GS}}=\lambda_t^{\mathrm{gas}}F_t^{\mathrm{GG}}+\sum_{g\in\mathcal{W}}c_{g,t}^{\mathrm{gas}}F_{g,t} \tag{1c}$$

$$C_t^{\mathrm{HS}}=\sum_{g\in\mathcal{M}\cup\mathcal{B}}c_{g,t}^{\mathrm{heat}}H_{g,t} \tag{1d}$$

where the operation cost for PSO C_t^{PS} comprises the energy trading cost with the upstream power grid as well as the power generation cost. The first two terms refer to buying active and reactive power from the upstream power grid, while the third and fourth terms correspond to the revenue from selling active and reactive power. The fifth and sixth terms illustrate the costs associated with the generation of active and reactive power, respectively. In the case of GSO, the operation cost C_t^{GS} is constituted by the trading cost with the upstream gas grid (1st term) and the gas generation cost (2nd term). For HSO, the operation cost C_t^{HS} is solely made up of the scheduling cost of heat generation units.

The mathematical models of every energy system within the LEC are detailed as follows.

1) Power System:

Power Generators. Small-scale diesel generators (DGs) and gas turbines (GTs) are prevalent in LECs as a power resource due to their characteristic of flexibility and reliability. Moreover, GT is a kind of energy conversion component that couples the gas network and power network. The operation boundaries for DGs and GTs are modeled as follows.

$$\underline{P}_g\le P_{g,t}\le\overline{P}_g,\forall g\in\mathcal{G}\cup\mathcal{T},\forall t\in T \tag{2a}$$

$$\underline{Q}_g\le Q_{g,t}\le\overline{Q}_g,\forall g\in\mathcal{G}\cup\mathcal{T},\forall t\in T \tag{2b}$$

$$|Q_{g,t}|\le P_{g,t}\tan\left(\arccos\sigma_g\right),\forall g\in\mathcal{G}\cup\mathcal{T},\forall t\in T \tag{2c}$$

$$P_{g,t}=F_{g,t}^{\mathrm{gt}}/\eta_g^{\mathrm{gt}},\forall g\in\mathcal{T},\forall t\in T \tag{2d}$$

where constraints (2a) and (2b) imply that the active and reactive outputs from the generator g are limited by the upper and lower boundaries $\underline{P}_g$, $\overline{P}_g$, $\underline{Q}_g$, and $\overline{Q}_g$. Constraint (2c) indicates the relationship between active and reactive power output of the generator g considering the impact of rated power factor σ_g. Constraint (2d) implies the energy transformation relationship between power output and gas supply of the GT unit g with energy conversion efficiency η_g^{gt}.

Energy Storage (ES). ES is a flexible option that is capable of handling power load fluctuation and hosting

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renewable energy through reasonable charging/discharging schemes. Considering the charging/discharging efficiency, the state of charge (SoC) dynamics of an ES can be modeled as:

$$S^{\mathrm{es}}_{m,t+1} = S^{\mathrm{es}}_{m,t} + \frac{P^{\mathrm{ch}}_{m,t}\Delta t \eta^{\mathrm{esc}}_m}{E^{\mathrm{es}}_m} + \frac{P^{\mathrm{dis}}_{m,t}\Delta t}{\eta^{\mathrm{esd}}_m E^{\mathrm{es}}_m}, \quad \forall m \in \mathcal{C}^p, \forall t \in T \tag{3a}$$

$$\underline{S}^{\mathrm{es}}_m \le S^{\mathrm{es}}_{m,t} \le \overline{S}^{\mathrm{es}}_m, \forall m \in \mathcal{C}^p, \forall t \in T \tag{3b}$$

$$0 \le P^{\mathrm{ch}}_{m,t} \le \overline{P}^{\mathrm{es}}_m \mathcal{I}^{\mathrm{es}}_{m,t}, \forall m \in \mathcal{C}^p, \forall t \in T \tag{3c}$$

$$\overline{P}^{\mathrm{es}}_m \left(\mathcal{I}^{\mathrm{es}}_{m,t} - 1\right) \le P^{\mathrm{dis}}_{m,t} \le 0, \forall m \in \mathcal{C}^p, \forall t \in T \tag{3d}$$

where constraint (3b) indicates the SoC dynamics of an ES with energy capacity E^{es}_m considering the charging efficiency η^{esc}_m and discharging efficiency η^{esd}_m. Constraint (3b) indicates the limited boundary for ESs' SoC. Constraints (3c) and (3d) demonstrate the upper and lower bounds of charging and discharging power with a binary variable $\mathcal{I}^{\mathrm{es}}_{m,t}$ distinguishing the status of the ES between charging ($\mathcal{I}^{\mathrm{es}}_{m,t} = 1$) and discharging ($\mathcal{I}^{\mathrm{es}}_{m,t} = 0$).

Power Network. The flow dynamics within the power network is formulated by a linearized branch flow model [24].

$$\sum_{i:i\to j} P_{ij,t} + p^G_{j,t} = \sum_{k:j\to k} P_{jk,t} + p^D_{j,t}, \forall j \in \mathcal{N}^p, \forall t \in T \tag{4a}$$

$$\sum_{i:i\to j} Q_{ij,t} + q^G_{j,t} = \sum_{k:j\to k} Q_{jk,t} + q^D_{j,t}, \forall j \in \mathcal{N}^p, \forall t \in T \tag{4b}$$

$$V_{j,t} = V_{i,t} - \frac{r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}}{U_0}, \forall ij \in \mathcal{L}^p, \forall t \in T \tag{4c}$$

$$P^2_{ij,t} + Q^2_{ij,t} \le \overline{S}^2_{ij,t}, \forall ij \in \mathcal{L}^p, \forall t \in T \tag{4d}$$

$$\underline{V} \le V_{j,t} \le \overline{V}, \forall j \in \mathcal{N}^p, \forall t \in T \tag{4e}$$

where the nodal active/reactive power balance of each power bus is implied in (4a) and (4b). Constraint (4c) describes the relationship of the nodal voltage between two adjacent power buses. Constraints (4d) and (4e) depict the operation limitation for line power flow and nodal voltage, which are constrained by line capacity $\overline{S}_{ij}$ and voltage limits $\underline{V}$, $\overline{V}$, respectively.

2) Gas System:

Gas Well (GW). GWs are the main supply sources of natural gas in the gas network. The gas output of GW g is modeled as follows, which is constrained by the upper and lower output boundaries $\underline{F}^{\mathrm{gw}}_g$, $\overline{F}^{\mathrm{gw}}_g$:

$$\underline{F}^{\mathrm{gw}}_g \le F_{g,t} \le \overline{F}^{\mathrm{gw}}_g, \forall g \in \mathcal{W}, \forall t \in T \tag{5a}$$

Gas Storage (GS). GS principally serves to address gas load variations by injecting gas during periods of low demand and withdrawing it during peak demand periods. The operation model of GS is similar to ES, which is modeled as follows.

$$S^{\mathrm{gs}}_{m,t+1} = S^{\mathrm{gs}}_{m,t} + \frac{F^{\mathrm{ch}}_{m,t}\Delta t \eta^{\mathrm{gsc}}_m}{E^{\mathrm{gs}}_m} + \frac{F^{\mathrm{dis}}_{m,t}\Delta t}{\eta^{\mathrm{gsd}}_m E^{\mathrm{gs}}_m}, \quad \forall m \in \mathcal{C}^g, \forall t \in T \tag{6a}$$

$$\underline{S}^{\mathrm{gs}}_m \le S^{\mathrm{gs}}_{m,t} \le \overline{S}^{\mathrm{gs}}_m, \forall m \in \mathcal{C}^g, \forall t \in T \tag{6b}$$

$$0 \le F^{\mathrm{ch}}_{m,t} \le \overline{F}^{\mathrm{gs}}_m \mathcal{I}^{\mathrm{gs}}_{m,t}, \forall m \in \mathcal{C}^g, \forall t \in T \tag{6c}$$

$$\overline{F}^{\mathrm{gs}}_m \left(\mathcal{I}^{\mathrm{gs}}_{m,t} - 1\right) \le F^{\mathrm{dis}}_{m,t} \le 0, \forall m \in \mathcal{C}^g, \forall t \in T \tag{6d}$$

where constraint (6a) indicates the SoC dynamics of a GS with gas capacity E^{gs}_m considering the charging efficiency η^{gsc}_m and discharging efficiency η^{gsd}_m. Constraint (6b) indicates the limited boundary for GSs' SoC. Constraints (6c) and (6d) demonstrate the upper and lower bounds of charging and discharging flux with a binary variable $\mathcal{I}^{\mathrm{gs}}_{m,t}$ distinguishing the status of the GS between charging ($\mathcal{I}^{\mathrm{gs}}_{m,t} = 1$) and discharging ($\mathcal{I}^{\mathrm{gs}}_{m,t} = 0$).

Gas Network. The safe operation of the gas network is secured by a model proposed in [25]:

$$\sum_{p:p\to b} F_{pb,t} + f^G_{b,t} = \sum_{n:b\to n} F_{bn,t} + f^D_{b,t}, \forall b \in \mathcal{N}^g, \forall t \in T \tag{7a}$$

$$\pi_{b,t} \le \pi_{p,t} \le \alpha_{bp}\pi_{b,t}, \forall bp \in \mathcal{L}^g_{\mathrm{act}}, \forall t \in T \tag{7b}$$

$$F^2_{bp,t} - \beta_{bp}\left(\pi^2_{b,t} - \pi^2_{p,t}\right) = 0, \forall bp \in \mathcal{L}^g_{\mathrm{ina}}, \forall t \in T \tag{7c}$$

$$\underline{\pi}_{b,t} \le \pi_{b,t} \le \overline{\pi}_{b,t}, \forall b \in \mathcal{N}^g, \forall t \in T \tag{7d}$$

$$\underline{F}_{bp} \le F_{bp,t} \le \overline{F}_{bp}, \forall bp \in \mathcal{L}^g, \forall t \in T \tag{7e}$$

where the nodal gas balance of gas node b is implied in (7a). Constraints (7b) and (7c) indicate the relationship of the nodal pressure between the two adjacent gas nodes in active/inactive gas pipelines, respectively. The pipelines without compressors are denoted as inactive pipelines $\mathcal{L}^g_{\mathrm{ina}}$, whereas those equipped with compressors are active pipelines $\mathcal{L}^g_{\mathrm{act}}$. Constraints (7d) and (7e) demonstrate the operation limitation for nodal pressure and gas flow.

3) Heat System:

Heat Generator. Electric heat pumps (EHPs) couple the power system and heat system for their capability of generating heat by consuming electricity, while gas boilers (GBs) serve as an alternative for meeting heat demand by consuming natural gas. The corresponding model of heat generators is formulated as:

$$H_{g,t} = P^{\mathrm{ehp}}_{g,t}\eta^{\mathrm{ehp}}_g, \forall g \in \mathcal{M}, \forall t \in T \tag{8a}$$

$$H_{g,t} = F^{\mathrm{gb}}_{g,t}\eta^{\mathrm{gb}}_g, \forall g \in \mathcal{B}, \forall t \in T \tag{8b}$$

$$0 \le H_{g,t} \le \overline{H}_g, \forall g \in \mathcal{M} \cup \mathcal{B}, \forall t \in T \tag{8c}$$

where constraint (8a) indicates the energy conversion from electricity to heat considering the conversion efficiency η^{ehp}_g. Constraint (8b) represents the energy conversion from gas to heat considering the conversion efficiency η^{gb}_g. The output boundary of energy conversion unit g is demonstrated in (8c).

Thermal Energy Storage (TS). TS can help handle the stochasticity from demand and increase the hosting capacity for renewable energy via smart charging/discharging. Following the same modeling fashion as ES and GS, the operation model of TS is formulated as:

$$S^{\mathrm{ts}}_{m,t+1} = S^{\mathrm{ts}}_{m,t} + \frac{H^{\mathrm{ch}}_{m,t}\Delta t \eta^{\mathrm{tsc}}_m}{E^{\mathrm{ts}}_m} + \frac{H^{\mathrm{dis}}_{m,t}\Delta t}{\eta^{\mathrm{tsd}}_m E^{\mathrm{ts}}_m}, \quad \forall m \in \mathcal{C}^h, \forall t \in T \tag{9a}$$

$$\underline{S}^{\mathrm{ts}}_m \le S^{\mathrm{ts}}_{m,t} \le \overline{S}^{\mathrm{ts}}_m, \forall m \in \mathcal{C}^h, \forall t \in T \tag{9b}$$

$$0 \le H^{\mathrm{ch}}_{m,t} \le \overline{H}^{\mathrm{ts}}_m \mathcal{I}^{\mathrm{ts}}_{m,t}, \forall m \in \mathcal{C}^h, \forall t \in T \tag{9c}$$

$$\overline{H}^{\mathrm{ts}}_m \left(\mathcal{I}^{\mathrm{ts}}_{m,t} - 1\right) \le H^{\mathrm{dis}}_{m,t} \le 0, \forall m \in \mathcal{C}^h, \forall t \in T \tag{9d}$$

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where constraint (9b) indicates the SoC dynamics of a TS with energy capacity E_m^{ts} considering the charging efficiency η_m^{tsc} and discharging efficiency η_m^{tsd}. Constraint (9b) indicates the limited boundary for TSs' SoC. Constraints (9c) and (9d) demonstrate the upper and lower bounds of charging and discharging power with a binary variable $\mathcal{I}_{m,t}^{\text{ts}}$ distinguishing the status of the TS between charging ($\mathcal{I}_{m,t}^{\text{ts}} = 1$) and discharging ($\mathcal{I}_{m,t}^{\text{ts}} = 0$).

Heat Network. The heat network is generally in a radial structure and consists of two types of pipelines, namely supply pipelines and return pipelines. Additionally, the heat network in this work operates under the quality regulation mode [26] that maintains a consistent mass flow rate. Considering the temperature mixing, heat loss, and heat exchange, the heat network is modeled as follows.

$$\sum_{i:i\to j} m_{ij} T_{ij,t}^{\text{out}} = T_{j,t}^{\text{mix}} \sum_{j:j\to k} m_{jk}, \forall j \in \mathcal{N}^h, \forall t \in T \tag{10a}$$

$$T_{ij,t}^{\text{end}} = \left(T_{ij,t}^{\text{start}} - T^{\text{amb}}\right) e^{-\frac{\kappa_{ij} d_{ij}}{cm_{ij}}} + T^{\text{amb}}, \quad \forall ij \in \mathcal{L}^s \cup \mathcal{L}^r, \forall t \in T \tag{10b}$$

$$H_{j,t}^G = cm_j\left(T_{j,t}^s - T_{j,t}^r\right), \forall j \in \mathcal{N}^s, \forall t \in T \tag{10c}$$

$$H_{j,t}^D = cm_j\left(T_{j,t}^s - T_{j,t}^r\right), \forall j \in \mathcal{N}^l, \forall t \in T \tag{10d}$$

$$\underline{T_j^s} \le T_{j,t}^s \le \overline{T_j^s} \tag{10e}$$

$$\underline{T_j^r} \le T_{j,t}^r \le \overline{T_j^r} \tag{10f}$$

where constraint (10a) describes the conservation of energy during the water mixing process in heat nodes. Constraints (10b) indicate the occurrence of heat loss within water pipelines. Constraints (10c) and (10d) describe the process of heat exchange that takes place in heat nodes with heat source/demand. The temperature boundaries of the supply and return water are demonstrated in (10e) and (10f), respectively.

B. MDP Formulation of Coordinated Dispatch

In essence, the coordinated LEC dispatch is a sequential stochastic optimization problem, which adheres to the Markov property that the future state of the system relies only on the current state and the selected action, without any connection to prior states; therefore, it can be formulated as an MDP which is characterized by a tuple $< \mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma >$, where $\mathcal{S}$ is the state space; $\mathcal{A}$ is the action space; $\mathcal{P}(s'|s, a, \chi) : \mathcal{S} \times \mathcal{A} \times \mathcal{X} \times \mathcal{S} \to \mathbb{R}$ is the probability of state transition considering the randomness $\mathcal{X}$ from demand and renewable energy; r is the immediate reward at each time step; $\gamma \in [0, 1]$ is the discount factor.

1) State: The overall state $\mathbf{s}_t$ mainly consists of three dependent states collected from PSO, GSO, and HSO.

- State of PSO $\mathbf{s}_t^{\text{PSO}}$ includes: the SoC of ESs $\mathbf{S}_t^{\text{es}}$, renewable energy output $\mathbf{P}_t^{\text{pv}}$ and $\mathbf{P}_t^{\text{wt}}$, active/reactive power demand $\mathbf{P}_t^D/\mathbf{Q}_t^D$, and the buying/selling prices of active power $\lambda_{\text{buy},t}^p/\lambda_{\text{sell},t}^p$. The prices of reactive power are not contained in the state because it is reasonable to assume that they follow the identical pattern as the prices of active power [27].
- State of GSO $\mathbf{s}_t^{\text{GSO}}$ includes: the SoC of GSs $\mathbf{S}_t^{\text{gs}}$ and gas demand $\mathbf{G}_t^D$.
- State of HSO $\mathbf{s}_t^{\text{HSO}}$ includes: the SoC of TSs $\mathbf{S}_t^{\text{ts}}$ and heat demand $\mathbf{H}_t^D$.

$$\mathbf{s}_t = \left\{\mathbf{s}_t^{\text{PSO}}, \mathbf{s}_t^{\text{GSO}}, \mathbf{s}_t^{\text{HSO}}, t\right\}, \forall \mathbf{s} \in \mathcal{S}, \forall t \in T \tag{11a}$$

$$\mathbf{s}_t^{\text{PSO}} = \left\{\mathbf{S}_t^{\text{es}}, \mathbf{P}_t^D, \mathbf{Q}_t^D, \mathbf{P}_t^{\text{pv}}, \mathbf{P}_t^{\text{wt}}, \lambda_{\text{buy},t}^p, \lambda_{\text{sell},t}^p\right\} \tag{11b}$$

$$\mathbf{s}_t^{\text{GSO}} = \left\{\mathbf{S}_t^{\text{gs}}, \mathbf{F}_t^D\right\} \tag{11c}$$

$$\mathbf{s}_t^{\text{HSO}} = \left\{\mathbf{S}_t^{\text{ts}}, \mathbf{H}_t^D\right\} \tag{11d}$$

2) Action: The overall action $\mathbf{a}_t$ is comprised of separate actions conducted by PSO, GSO, and HSO:

$$\mathbf{a}_t = \left\{\mathbf{a}_t^{\text{PSO}}, \mathbf{a}_t^{\text{GSO}}, \mathbf{a}_t^{\text{HSO}}\right\}, \forall \mathbf{a} \in \mathcal{A}, \forall t \in T \tag{12a}$$

$$\mathbf{a}_t^{\text{PSO}} = \left\{\mathbf{a}_t^{\text{dg},p}, \mathbf{a}_t^{\text{gt},p}, \mathbf{a}_t^{\text{dg},q}, \mathbf{a}_t^{\text{gt},q}, \mathbf{a}_t^{\text{es}}\right\} \tag{12b}$$

$$\mathbf{a}_t^{\text{GSO}} = \left\{\mathbf{a}_t^{\text{gw}}, \mathbf{a}_t^{\text{gs}}\right\} \tag{12c}$$

$$\mathbf{a}_t^{\text{HSO}} = \left\{\mathbf{a}_t^{\text{ehp}}, \mathbf{a}_t^{\text{gb}}, \mathbf{a}_t^{\text{ts}}\right\} \tag{12d}$$

where $\mathbf{a}_t^{\text{PSO}}$, $\mathbf{a}_t^{\text{GSO}}$, $\mathbf{a}_t^{\text{HSO}}$ correspond to dependent actions taken by PSO, GSO, and HSO. Actions of $\mathbf{a}_t^{\text{dg},p}, \mathbf{a}_t^{\text{gt},p}, \mathbf{a}_t^{\text{dg},q}, \mathbf{a}_t^{\text{gt},q}, \mathbf{a}_t^{\text{gw}}, \mathbf{a}_t^{\text{ehp}}, \mathbf{a}_t^{\text{gb}} \in [0, 1]$ denote the output rate for diverse energy generation units. The charging and discharging rates for different energy storages are denoted by $\mathbf{a}_t^{\text{es}}, \mathbf{a}_t^{\text{gs}}, \mathbf{a}_t^{\text{ts}} \in [-1, 1]$. It is worth noting that the actual value of the output is obtained via the product of the upper output limit of the energy unit and the corresponding output rate. Therefore, the power capacity constraint can be inherently fulfilled.

3) State Transition: In the problem of coordinated dispatch, the state transition from the current state $\mathbf{s}_t$ to the next state $\mathbf{s}_{t+1}$ is partially stochastic. Given an action $\mathbf{a}_t$, the SoC of ES, GS, and TS will transition to a deterministic state. However, the state transitions of renewable energy, energy prices, and energy demand are stochastic due to the existence of the unobservable environmental randomness χ.

4) Reward: An immediate reward r_t is allocated following the state transition of each time step, which is quantified as the negative summation of the operation cost generated by operators.

$$r_t = -\left(C_t^{\text{PS}} + C_t^{\text{GS}} + C_t^{\text{HS}}\right), \forall t \in T \tag{13}$$

III. Methodology

The proposed privacy-enhanced safe soft actor-critic (PSSAC) framework for coordinated dispatch is depicted in Fig. 2. A central server (CS), containing the actor-critic architecture of the RL-based coordination algorithm, is included within the PSSAC framework to conduct real-time coordinated dispatch for multiple system operators, including PSO, HSO, and GSO. Compared with conventional centralized RL-based dispatch frameworks, the proposed framework equips each operator with two additional modules: a perturbation module for privacy enhancement and a safety module for secure LEC operation. The workflow of the PSSAC framework can be outlined as follows: Initially, each operator conceals its private

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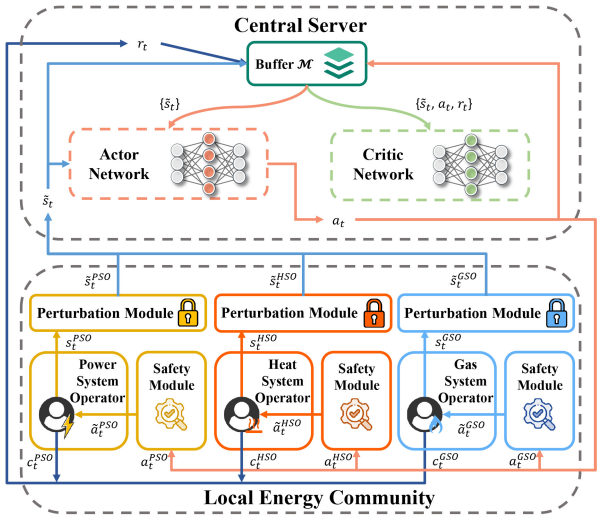


Fig. 2. The proposed PSSAC framework for coordinated dispatch.

state using the perturbation module and then uploads it to the CS. Subsequently, the CS outputs the coordination result based on the aggregated states and distributes the recommended operation actions to operators. Lastly, operators assess the recommended actions and correct them if necessary with the safety modules before the final execution.

A. Reinforcement Learning for Coordinated Dispatch

In this paper, we adopt the Soft Actor-Critic (SAC) algorithm [28] for achieving coordinated LEC dispatch, owing to its learning stability and high sampling efficiency. To learn a policy that explores the environment sufficiently while still achieving success at the task, SAC incorporates an entropy term, which measures the stochasticity of the policy, into the reward function to promote exploration.

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s_t}, \mathbf{a_t}) + \alpha\mathcal{H}(\pi(\cdot|\mathbf{s}_t))] \quad (14)$$

where ρ_π is the marginal probability for state-action tuples under the current policy, $\mathcal{H}(\cdot)$ is the entropy measure for the policy π, and α is the temperature parameter that implies the relative importance of the entropy term.

By alternating between optimizing a parameterized soft Q-function $Q_\theta(\mathbf{s}_t, \mathbf{a}_t)$ and a tractable policy function $\pi_\phi(\mathbf{a}_t|\mathbf{s}_t)$, the agent can learn to approach the optimal policy and maximize the cumulative reward. The training of the soft Q-function is designed to minimize the soft Bellman residual:

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t)\sim\mathcal{D}}\left[\frac{1}{2}\Big(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \hat{Q}(\mathbf{s}_t, \mathbf{a}_t)\Big)^2\right] \quad (15)$$

with

$$\hat{Q}(\mathbf{s}_t, \mathbf{a}_t) = \Big(r(\mathbf{s_t}, \mathbf{a_t}) + \gamma\mathbb{E}_{(\mathbf{s}_{t+1})\sim\rho_\pi(s)}\big[V(\mathbf{s}_{t+1})\big]\Big) \quad (16)$$

$$V(\mathbf{s}_t) = \mathbb{E}_{\mathbf{a}_t\sim\pi}\big(Q_{\overline{\theta}}(\mathbf{s_t}, \mathbf{a_t}) - \alpha\log\pi_\phi(\mathbf{a}_t|\mathbf{s}_t)\big) \quad (17)$$

where $V(\mathbf{s}_t)$ denotes the state value of state $\mathbf{s}_t$, $\overline{\theta}$ denote the parameters of the target soft Q-function.

The policy is updated by minimizing the Kullback-Leibler divergence between the current policy and the exponential of the new soft Q-function, which guarantees that the policy can be improved regarding its soft value.

$$\pi^{\text{new}} = \arg\min_{\pi'\in\Pi} D_{KL}\left(\pi'(\cdot|\mathbf{s}_t)\|\frac{\exp\Big(Q^{\pi^{\text{old}}}(\mathbf{s}_t, \cdot)\Big)}{Z^{\pi^{\text{old}}}(\mathbf{s}_t)}\right) \quad (18)$$

where Π is the set of potential policies, $Z^{\pi^{\text{old}}}(\mathbf{s}_t)$ is the partition function for normalizing the distribution. The corresponding objective function for policy update can be written as:

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t\sim\mathcal{D}}\big[\mathbb{E}_{\mathbf{a}_t\sim\pi_\phi}\big[\alpha\log\big(\pi_\phi(\mathbf{a}_t|\mathbf{s}_t)\big) - Q_\theta(\mathbf{s}_t, \mathbf{a}_t)\big]\big] \quad (19)$$

where $\mathcal{D}$ denotes the replay buffer.

As the optimal temperature parameter varies across tasks and during different training periods, updating the temperature parameter during training can lead to a more stable algorithm performance than using a fixed α. The optimal temperature parameter is learned by minimizing the objective function (20), which aims to enforce a minimum entropy constraint on the policy during policy improvement:

$$J(\alpha) = \mathbb{E}_{\mathbf{a}_t\sim\pi}\big[-\alpha\log\pi(\mathbf{a}_t|\mathbf{s}_t) - \alpha\mathcal{H}_0\big] \quad (20)$$

where $\mathcal{H}_0$ is the predefined minimum threshold of the policy entropy.

It should be noted that the output of the SAC actor-network is confined by the application of a tanh function, which ensures that each component of the output action vector falls within the range of $[-1, 1]$. The operations of storage devices can be conveniently symbolized with positive or negative values for discharging or charging rates, which are contained within the $[-1, 1]$ range. The actions for energy generation devices are converted to fit within the [0, 1] range by utilizing the mapping function $a' = \frac{a+1}{2}$.

B. Perturbation Module for Privacy Enhancement

The formulated MDP in Section II-B reveals that the states contain sensitive information about each operator, such as energy demand and the SOC for energy storage. This information can expose the hourly operational status of the operator, leading to potential privacy breaches. To protect the privacy of system operators, it is essential to conceal each operator's private state prior to uploading it to the CS. Considering the compatibility with RL training and the inherent heterogeneity among operators' states, the proposed perturbation mechanism needs to provide differentiated perturbation for different operators while preserving the spatial characteristics of the original state space.

Drawing inspiration from multiplicative perturbation [29], we propose a privacy-enhanced perturbation mechanism for the RL-based coordinated dispatch problem. The proposed perturbation procedure can be described as follows. Before the state is uploaded, each operator independently applies differentiated geometric operations (rotation and translation) on the initial state. This process transforms the original state points into new ones, making it challenging to ascertain the initial states. Consequently, the sensitive information within the state is disguised.

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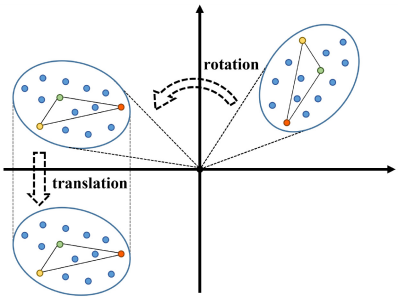


Fig. 3. Diagram of the privacy-enhanced mechanism.

Specifically, we secure the private state by incorporating rotation and translation perturbations to the original state $\mathbf{s}_t^o$ of the operator o, ultimately resulting in the formation of the concealed state $\tilde{\mathbf{s}}_t^o$.

$$\tilde{\mathbf{s}}_t^o = \mathbf{R}_o \mathbf{s}_t^o + \Psi_o \tag{21}$$

The rotation matrix $\mathbf{R}_o$ and the translation vector Ψ_o are different across operators, which act as the algebraic encryption keys (EKs). Moreover, the matrix $\mathbf{R}_o$ is an orthonormal matrix, which can be generated by following the Haar distribution [30].

$$\mathbf{R}_o^T \mathbf{R}_o = \mathbf{R}_o \mathbf{R}_o^T = \mathbf{I} \tag{22}$$

To this end, the proposed state perturbation mechanism is an isometric transformation, i.e., the relative distances between any two states $\mathbf{x}$, $\mathbf{y}$ of a system operator are maintained during the process of perturbation.

$$\begin{aligned} \|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\| &= \|\mathbf{R}_o \mathbf{x} + \Psi_o - \mathbf{R}_o \mathbf{y} - \Psi_o\| \\ &= \mathbf{x}^T \mathbf{R}_o^T \mathbf{R}_o \mathbf{x} - \mathbf{x}^T \mathbf{R}_o^T \mathbf{R}_o \mathbf{y} \\ &\quad - \mathbf{y}^T \mathbf{R}_o^T \mathbf{R}_o \mathbf{x} + \mathbf{y}^T \mathbf{R}_o^T \mathbf{R}_o \mathbf{y} \\ &= \|\mathbf{x} - \mathbf{y}\| \end{aligned} \tag{23}$$

The perturbation process in a 2D plane is straightforward to visualize, which is shown in Fig. 3. As depicted in the diagram, the shape of the triangle, which represents the relative distance between three state points, remains unchanged throughout the perturbation process. This observation highlights that the isometric perturbation mechanism conceals private information while preserving the inner structure of the state space. Each operator performs the perturbation process with its independent perturbation module. Subsequently, the CS constructs a comprehensive input state $\tilde{\mathbf{s}}_t = \{\tilde{\mathbf{s}}_t^{\text{PSO}}, \tilde{\mathbf{s}}_t^{\text{GSO}}, \tilde{\mathbf{s}}_t^{\text{HSO}}, t\}$ by concatenating the uploaded concealed states from all energy system operators.

Although operators conceal their own states using distinct rotation matrices and translation vectors, the isometry characteristic is preserved in the concatenated concealed state space, as distance calculations are performed feature-wise. Leveraging this property, we further develop the categorized perturbation to bolster the mechanism's security in the context of large state spaces. Specifically, an operator's state features can be categorized into groups according to any chosen criteria, and subsequently concealed on a group-by-group basis. Consequently, the computational difficulty of reversing the transformation progressively increases as the number of groups escalates. We take the categorized perturbation of 2 subgroups as an example. Given the components of an original state vector are divided into 2 subgroups, i.e., $\mathbf{s}_t^o = [\mathbf{g_1}, \mathbf{g_2}]$, the concealed state will be $\tilde{\mathbf{s}}_t^o = [\mathbf{R}_o^{g1}\mathbf{g}_1 + \Psi_o^{g1}, \mathbf{R}_o^{g2}\mathbf{g}_2 + \Psi_o^{g2}]$.

In the proposed perturbation mechanism, we did not consider the need for the regular change of EKs, which is crucial in mitigating potential harm in the event of a key being compromised. However, our proposed method can handle this need with moderate modification, specifically by the introduction of a mapping key that specifies the correspondence between the new concealed state space and the concealed training space. Furthermore, policy distillation [31] is also a potential solution for transitions in EKs, which highlights that a trained reinforcement learning model (using old EKs) can be used to guide another model (with changed EKs) in swiftly adapting to a new state space. To summarize, the regular rotation of EKs is a significant practice for enhancing security, with several viable approaches to achieving this goal. Within this context, a thorough and formal investigation into the transition of EKs will be reserved for future study.

C. Security Analysis for Perturbation Module

In this section, we evaluate the security of the proposed perturbation module by analyzing a set of potential attacks with different levels of additional knowledge about the original states. The proposed perturbation mechanism shares the privacy guarantee with multiplicative perturbation methods. One significant difference is that the distribution of the original state data changes as the interaction between the operators and CS is continuous, which makes it more challenging for the attacks to reverse the transformation. The following analysis is based on a frozen state set, which means we freeze the interaction process, and look at the states stored and processed on the CS platforms.

Naive Estimation. This kind of attack estimates the original states only from the concealed states with no additional knowledge. Align with [29], the privacy guarantee of the state $\mathbf{s}_t^o$ is measured with the variance-of-difference (VoD) metric $VoD_t = Var(\tilde{\mathbf{s}}_t^o - \mathbf{s}_t^o)$ which reflects the complexity involved in estimating the original state. Considering the whole frozen state set, VoD_t is evaluated by

$$\begin{aligned} Cov\left(\tilde{\mathbf{S}^o} - \mathbf{S}^o\right)_{(t,t)} &= Cov\left(\mathbf{R}_o \mathbf{S}^o - \mathbf{S}^o\right)_{(t,t)} \\ &= \sum_{j=1}^{d}\sum_{k=1}^{d} r_{tj} r_{tk} c_{kj} - 2\sum_{j=1}^{d} r_{tj} c_{tj} + c_{tt} \end{aligned} \tag{24}$$

where $\mathbf{S}^o$ and $\tilde{\mathbf{S}^o}$ are the set of actual states and the set of concealed states of the operator o, d is the dimension of the state vector defined in the formulated MDP, r_{tj} and c_{tj} are the element (t, j) in the matrix $\mathbf{R}_o$ and the covariance matrix of $\mathbf{S}^o$, respectively. Considering that the rotation matrix is generated following the Haar distribution, it is reasonable to assume all elements within the rotation matrix approximately adhere to

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an independent normal distribution $N(0, 1/d)$, based on which the mean and variance of the VoD_t are derived as follows.

$$E[VoD_t] \sim c_{tt} \tag{25}$$

$$Var(VoD_t) \sim O\left(1/d^2 \sum_{j=1}^{d}\sum_{k=1}^{d} c_{kj}^2 + 4/d \sum_{j=1}^{d} c_{tj}^2\right) \tag{26}$$

Based on the above equations, we can find that the basic privacy guarantee of the proposed method is determined by the variance of the original state and the average of the squared covariance entries of the state set. As suggested in [32], rotation perturbation offers sufficient privacy guarantee in the case that no additional knowledge is known by the attacker, and a higher level of privacy guarantee can be achieved by optimizing the order of the rows in the rotation matrix $\mathbf{R}_o$. Moreover, the introduction of the translation vector improves the privacy guarantee of the vulnerable points close to the origin.

ICA-Based Attacks. Due to the intrinsic permutation ambiguity feature of Independent Component Analysis (ICA) [33], the success of the ICA-based attacks relies on the knowledge of the original state set's distribution. Mathematically, this can be demonstrated by the fact that any invertible permutation matrix $\mathbf{P}$ can be applied to the model, resulting in

$$\tilde{\mathbf{S}}^o = \mathbf{R}_o\mathbf{P}^{-1}\mathbf{P}\mathbf{S}^o \tag{27}$$

where $\tilde{\mathbf{S}}^o$ and $\mathbf{S}^o$ are the concealed state set and original state set, respectively. Therefore, without prior knowledge of the original state distributions, ICA might only yield an estimation of a permuted state set $\mathbf{P}\mathbf{S}^o$.

Distance-Inference Attacks. To conduct this kind of attack, the attacker requires the knowledge of a set of original states. Based on the known state points, it is possible to match them in the set of concealed states by a simple exhaustive search even in a high-dimension case. However, the cost of the exhaustive search is considerable. Once the match is done, there are two scenarios for the subsequent attack on other original states. If the known state points are linearly independent and their number is at least $d+1$, where d is the dimension of the state vector defined in the formulated MDP, the rotation matrix and translation vector can be precisely calculated. This implies that the perturbation process has been fully deciphered, rendering other original states no longer secure. However, if the number of known points is less than d, the estimate of the rotation matrix $\tilde{\mathbf{R}}_o$ is not unique. In this case, the privacy can still be preserved to a certain extent.

To summarize, our proposed framework ensures a secure exchange of information by only sharing concealed states with the CS, while maintaining the original operation states locally, thus preventing public disclosure. This lack of access to additional information about the actual states significantly diminishes the effectiveness of ICA-based and distance-inference attacks. Furthermore, an optimized rotation matrix can achieve a high level of privacy guarantee against naive estimation. Consequently, the proposed perturbation module can be considered secure and capable of preserving privacy. Additionally, the proposed categorized perturbation can further enhance security, which groups sensitive information and conducts perturbation independently. This implies that any potential attacker aiming to successfully decode the perturbation would necessitate not only an accurate restoration of the algebraic EKs but also comprehensive knowledge of the precise grouping scheme.

D. Safety Module for Safe Operation

Prior to the ultimate implementation of the scheduling decision, each operator employs its dedicated safety module to evaluate the recommended action distributed by the CS and make adjustments if necessary. The core of the safety module is to solve an optimization problem that aims at minimizing the action changes while ensuring safe operation:

$$\begin{aligned} \min_{\tilde{\mathbf{a}}_{o,t}} \quad & \|\tilde{\mathbf{a}}_{o,t} - \mathbf{a}_{o,t}\|^2 \\ \text{s.t.} \quad & f\left(\tilde{\mathbf{a}}_{o,t}\right) \le 0 \end{aligned} \tag{28}$$

where $\tilde{\mathbf{a}}_{o,t}$ is the safe action of the operator o, $\mathbf{a}_{o,t}$ is the action distributed by the CS, and $f(\cdot)$ refers to the system operational constraints in equation (3b)/(4)/(6b)/(7)/(9b)/(10).

The safety module functions not only in the implementation phase but also in the training phase. The importance of safety during training is underscored by two primary reasons. First, unsafe actions often lead to failure states, e.g., overcharging and overdischarging of energy storage, generating adverse experiences that can decelerate policy improvement and even impede convergence. Ensuring safety during training helps to prevent these failures, thereby enhancing the efficiency and stability of the learning process. Moreover, the integration of the safety module can be viewed as a modification of the environment, which alters the dynamics of the environment. The minimization of the operational cost is achieved during the training process, where the RL agent operates through trial and error within the modified environment, thereby learning an optimal policy from its interaction experiences.

Due to the existence of the energy conversion units, the energy supply for one energy system (e.g., heat system) may lead to an energy load on another system (e.g., power system and gas system). To this end, the safety modules would communicate with each other during the correction process. These modules can work under the framework of distributed optimization to reach the optimal solution. However, to enhance computational efficiency, this paper strategically executes the correction of safety modules in a sequential manner.

It is worth noting that the security and privacy of the system model can be well maintained since the safety module resides on the operator's side. As a result, no private information about the system model will be disclosed to the CS.

E. Full Algorithm

There are two main issues that we aim to solve with the proposed framework, which are privacy enhancement and safe operation respectively. To achieve this goal, we incorporate two modules, namely the perturbation module and the safety module, into the interaction loop between CS and operators. Through continuous interaction and improvement, the agent

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can learn to achieve safe coordinated dispatch in the concealed state space.

In the proposed method, we adopt one policy network, two Q-networks, and two target Q-networks. The two Q-networks are trained independently, and the policy network is updated using the minimum of the parameterized Q-networks. In this way, the positive bias during policy improvement is mitigated, and the training process is accelerated.

The training process of the proposed method contains two main stages that correspond to experience accumulation and network update respectively. In the stage of experience accumulation, CS firstly samples action $\mathbf{a}_t$ based on the concealed states $\tilde{\mathbf{s}}_t$ collected from operators (LINE 4-5). The sampled action will then be divided and distributed to the corresponding operator (LINE 6). Through the assessment and correction process of individual safety modules, operators obtain the safe action $\tilde{\mathbf{a}}_{o,t}$ and use it for operation (LINE 7-8). Once the operational action is finalized, each operator can locally compute the operation cost and ascertain the subsequent state. Both the operation cost and the subsequent state are then transmitted to the CS, with the state being concealed prior to transmission. The CS consolidates the operation cost to yield the reward r_t and concatenates the uploaded states from operators to establish the subsequent concealed state $\tilde{\mathbf{s}}_{t+1}$ (LINE 9). Until then, experience accumulation for one episode is finished, and the experience $\{\tilde{\mathbf{s}}_t, \mathbf{a}_t, \tilde{\mathbf{s}}_{t+1}, r_t\}$ is stored in the replay buffer $\mathcal{D}$ (LINE 10). In the stage of network update, the policy network, Q-networks, and the temperature parameter are improved with a batch of sampled experiences from the replay buffer (LINE 12-15). The weights of the target networks $\overline{\theta}_i$ are updated in a different manner, which are obtained as an exponentially moving average of the Q-network weights θ_i (LINE 16). The pseudo-code of the proposed PSSAC algorithm is shown in Algorithm 1.

Algorithm 1 PSSAC

1: Initialize: & θ_1, θ_2, ϕ,
$\overline{\theta}_1 \leftarrow \theta_1$, & $\overline{\theta}_2 \leftarrow \theta_2$, $\mathcal{D} \leftarrow \emptyset$
2: **for** episode e=1 to E **do**
3: **for** time step t=1 to T **do**
4: Collect the concealed state from operators and form the concatenated state $\tilde{\mathbf{s}}_t$
5: Sample action $\mathbf{a}_t$ according to policy $\pi_\phi(\mathbf{a}|\mathbf{s})$
6: Distribute operation action $\mathbf{a}_{o,t}$ to each operator
7: Obtain safe operation action $\tilde{\mathbf{a}}_{o,t}$ from individual safety module
8: Execute safe operation action by each operator
9: Observe the next state $\tilde{\mathbf{s}}_{t+1}$ and reward r_t
10: Store the transition $\{\tilde{\mathbf{s}}_t, \mathbf{a}_t, \tilde{\mathbf{s}}_{t+1}, r_t\}$ to the replay buffer $\mathcal{D}$
11: **end for**
12: Sample a batch of experience from the replay buffer $\mathcal{D}$
13: Update soft Q-network weights θ_i according to Eq. (15)
14: Update policy network weights ϕ according to Eq. (19)
15: Update temperature parameter α according to Eq. (20)
16: Update target soft Q-network weights as an exponentially moving average of θ_i, i.e., $\overline{\theta}_i \leftarrow \tau\theta_i + (1-\tau)\overline{\theta}_i$ for $i \in \{1, 2\}$
17: **end for**

IV. Case Studies

A. Experimental Setups

Fig. 4 shows the topology of the experimental LEC network modified from [34], which consists of a 6-bus power network, a 7-node gas network, and an 8-node heat network. Besides, the LEC network contains ten distinct energy components, including 1 PV, 1 WT, 1 DG, 1 GT, 1 ES, 2 GWs, 1 GS, 1 GB, 1 EHP, and 1 TS. The operational data of power demand, heat demand, and renewable energy generation are sourced from the UK Nation Grid [35]. The data of gas demand is sourced from [34]. In this work, we adopt a time-of-use (TOU) electricity pricing scheme, with details presented in Table I. The cost of purchasing or generating reactive power is set to be equivalent to 10% of the corresponding cost associated with active power. The cost coefficient of energy units is presented in Table II. Meanwhile, we assume a constant natural gas price of 0.2 £/Sm^3. The constructed dataset spans a continuous period of one year, the statistical characteristic of which is shown in Table III. Additionally, we allocate the first 11 months as the training set and the final month as the test set.

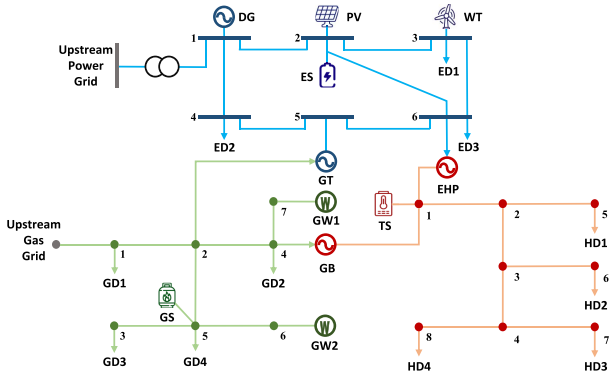


Fig. 4. Diagram of the experimental LEC network.

TABLE I
Time-of-Use Electricity Price

Type	Time Slot	Buying Price (£/kWh)	Selling Price (£/kWh)
Normal	10:00 - 16:00; 21:00 - 23:00	0.10	0.05
Peak	7:00 - 10:00; 16:00 - 21:00	0.24	0.12
Off-Peak	23:00 - 7:00	0.04	0.02

To validate the performance of the proposed method, five algorithms are used as the baselines for comparison, including perfect Mixed-Integer Linear Programming (MILP), safe SAC (SSAC), penalty-based SAC, stochastic MPC, stochastic MILP.

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TABLE II
THE COST COEFFICIENT OF ENERGY UNITS

	DG	GT	GW	EHP	GB
Cost Coefficient	0.21 (£/kWh)	0.08 (£/kWh)	0.20 (£/Sm^3)	0.05 (£/kWh)	0.04 (£/kWh)

TABLE III
THE STATISTIC CHARACTERISTICS OF THE DATASET

	Max Value	Min Value	Average Value
Power (kW)	310.30	64.05	170.45
Heat (kW)	278.20	0.55	73.26
Gas (Sm^3)	341.50	160.24	258.12
WT (kW)	50.00	0	11.01
PV (kW)	160.00	0	32.38

TABLE IV
HYPERPARAMETERS OF THE ALGORITHM

Symbol	Hyperparameter	Value
N_{batch}	Batch size	512
$N_{\mathcal{D}}$	Replay buffer size	720000
N_l	Number of hidden layers	2
N_u	Number of hidden units	256
τ	Soft update factor	5×10^{-3}
γ	Discount facotr	0.95
η_{actor}	Actor learning rate	10^{-4}
η_{critic}	Critic learning rate	10^{-4}
τ	Target smoothing coefficient	5^{-3}
$\mathcal{H}_0$	Entropy target	$-\dim(\mathcal{A})$
k_1	Weight of operation cost	1
k_2	Weight of penalty cost	0.1
	Optimizer	Adam

1) Perfect MILP: With perfect information about the system model and future uncertainties, coordinated dispatch is addressed by solving a deterministic MILP problem.

2) SSAC: The SSAC algorithm serves as an intermediate version of the proposed method. In this approach, we equip the vanilla SAC algorithm with a safety module to prevent unsafe exploration, while not considering privacy security.

3) Penalty-Based SAC: To assist the agent in learning to perform secure operations, a penalty term for constraint violation is integrated into the reward function of the vanilla SAC algorithm. The reward function is in the form as follows.

$$r_t = -\left[k_1\left(C_t^{\mathrm{PS}} + C_t^{\mathrm{GS}} + C_t^{\mathrm{HS}}\right) + k_2 C_t^{\mathrm{pen}}\right], \forall t \in T \quad (29)$$

$$C_t^{\mathrm{pen}} = \sum_{i\in\Omega} \mathrm{CV}_t(i), \forall t \in T \quad (30)$$

where C_t^{pen} is the penalty cost for the constraint violation. k_1 and k_2 are weights indicating the relative importance of minimizing the operation costs versus the penalty cost. $\mathrm{CV}_t(i)$ is the violation measure of constraint i at time step t, e.g., demand unbalance, voltage violation, etc. Ω is the operational constraint set of the LEC system that includes constraints (3b), (4), (6b), (7), (9b), and (10).

4) Stochastic MPC: A stochastic MPC optimization problem is solved in a rolling manner (i.e., on an hourly basis), taking into account the near-future information (forecast data for the upcoming 4 hours).

5) Stochastic MILP: Taking into account the uncertainties inherent in the forecast data, a scenario-based stochastic MILP problem is formulated and resolved for daily coordinated dispatch of an LEC.

Note that the demand data and renewable data collected from [34] and [35] are used as real values. However, operators have no access to the real values during the decision-making phase. Instead, they make decisions based on imperfect information with uncertainties, e.g., predicted values. While the performance metrics (e.g., operation cost) are assessed when the real values are revealed. Therefore, uncertainties within demand and renewable data need to be considered and simulated in the experiments. In this study, the uncertainties (including electricity demand, heat demand, gas demand, wind generation, and PV power generation) are accommodated by utilizing normal distributions. For scenario-based approaches, such as Stochastic MPC and Stochastic MILP, the perfect value is regarded as the mean, with the standard deviation (Stdv.) being 10% of this value. A total of 100,000 scenarios are uniformly sampled from the normal distributions. The scenario reduction technique in [36] is used to identify the most representative scenario sets. As for the data-driven approach, such as SSAC, Penalty-based SAC, and PSSAC, we generate the training and test sets similarly. The data in these sets are generated from a normal distribution, where the mean is the perfect value and the Stdv. is 10% of this mean.

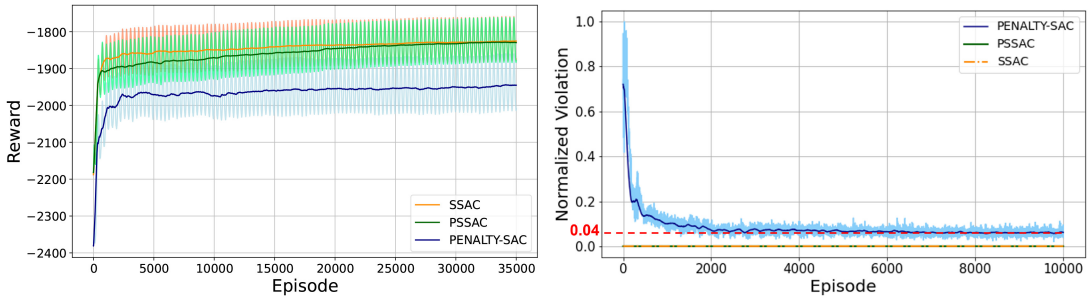
The hyperparameters of the SAC algorithm are listed in Table IV. All experiments are carried out on a workstation with NVIDIA GeForce RTX 3080 Ti GPU, Intel Xeon W-3335 CPU, and 188GB RAM. All codes are written based on Python language and Pytorch framework.

B. Algorithm Training Performance

To verify the effectiveness of the proposed algorithm, we conduct an ablation experiment in terms of the safety module and the perturbation module, involving the algorithms of Penalty-based SAC, SSAC, and PPSAC. Due to the limited length of operators' state vectors, the private states are concealed in one shot instead of using categorized perturbation. The training result is shown in Fig. 5.

Figure 5(a) presents the convergence outcomes for the three distinct algorithms. Lighter-colored curves signify the episodic rewards obtained throughout the training process, while darker-colored curves denote the corresponding moving average values. It is important to note that reward curves continue to fluctuate periodically even after convergence, primarily due to the time-dependent nature of the dataset, particularly regarding energy demand. Compared to the penalty-based SAC, both SSAC and PSSAC exhibit significantly higher starting points and final convergence values. This can be attributed to the safety module's ability to guide the agent toward safe exploration while avoiding penalties caused by constraint violations. Moreover, both PSSAC and SSAC converge to a comparable level within 5,000 episodes. After that, a 'tiny improvement' phenomenon occurs: SSAC stabilizes at

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(a) Training reward (b) Normalized constraint violation

Fig. 5. The training performance of the algorithms in terms of training reward and normalized constraint violation.

a marginally higher reward, whereas PSSAC demonstrates progressive improvement, ultimately attaining the same episodic reward level as SSAC after 23,000 episodes. This observation suggests that, although the perturbation module may introduce additional complexity to the exploration process, the training performance of the algorithm can still be maintained.

The constraint violation results for the algorithms are depicted in Fig. 5(b). The light blue curve represents the normalized constraint violation of the penalty-based SAC, while the dark blue curve displays its corresponding moving average values. The green curve and the dotted orange curve illustrate the violation outcomes for PSSAC and SSAC, respectively. As seen in this figure, the normalized constraint violation of the penalty-based SAC diminishes throughout the training and ultimately converges at a value of approximately 0.04. In contrast, the proposed PSSAC and SSAC exhibit no constraint violation during the entire training process, highlighting the effectiveness of the safety module.

C. LEC Operational Results

In this section, we evaluate the proposed PSSAC algorithm and the baseline algorithms on the test set. All neural network-based algorithms, specifically PSSAC, SSAC, and penalty-based SAC, are initially trained under the same settings of hyperparameters. Subsequently, each algorithm performs coordinated dispatch over a span of 31 test days.

Fig. 6 displays the daily operational cost curves for the various algorithms on the test set, while Fig. 7 presents a comparison of all algorithms in terms of their cumulative operational costs and the corresponding ratios relative to the perfect MILP solution. It can be observed that the daily cost curves of PSSAC and SSAC nearly coincide with each other, displaying the same trend as the optimal solution curve. Additionally, they are the closest to the optimal solution in terms of distance, with the smallest gap (cumulatively 3% difference), indicating that the proposed method and SSAC outperform in handling uncertainty and reducing costs. It also verified that the perturbation process almost has no impact on the test performance of the algorithm. On the contrary, the daily cost curves of the remaining baseline algorithms display significant fluctuation patterns compared to the optimal solution curve. This ultimately leads to a considerable gap, with more than 9% difference in cumulative cost from perfect

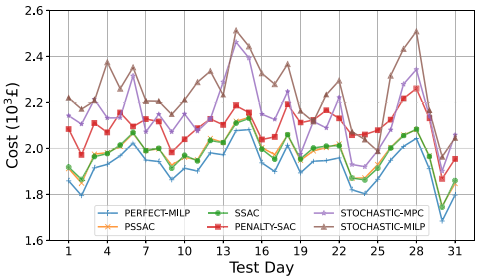


Fig. 6. The curves of the daily cost for different algorithms on the test set.

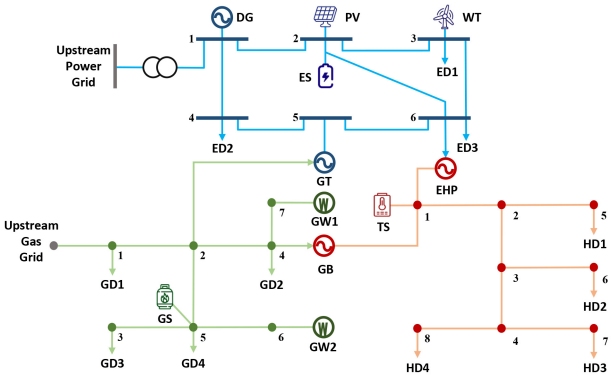


Fig. 7. The cumulative cost of different algorithms on the test set.

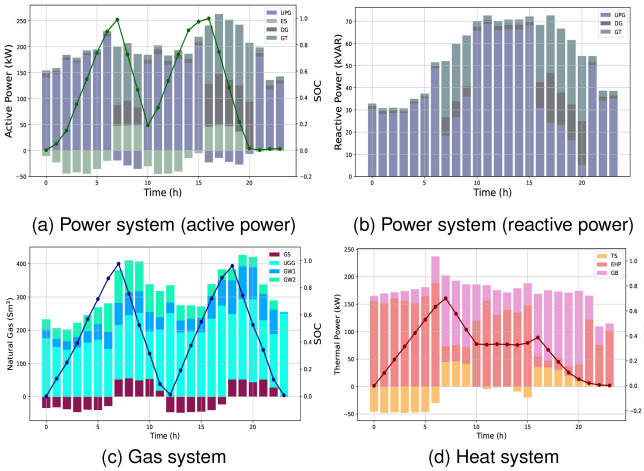
MILP. The inferior performance of the penalty-based SAC and stochastic algorithms can be attributed to two main factors. Firstly, the penalty-based SAC may execute operational actions that violate network constraints during the scheduling process, resulting in additional penalty costs. Secondly, the stochastic-MPC and stochastic-MILP rely heavily on forecast data of future time steps, making their performance easily susceptible to inaccurate forecast information.

We also tested the algorithms' performance under different degrees of system uncertainties by adjusting the Stdv. of the preset normal distributions. The result is shown in Table V. Under varying degrees of uncertainty, we can observe that the performance of RL-based methods remains relatively stable. Conversely, the performance of stochastic methods is highly sensitive to changes in system uncertainty. In cases of 5% uncertainties, the performance of stochastic MPC and stochastic MILP outperform Penalty-SAC by 2.51% and 0.95%, respectively. However, there is still a gap between the

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TABLE V
THE TEST PERFORMANCE OF PSSAC AND BASELINES UNDER DIFFERENT UNCERTAINTIES

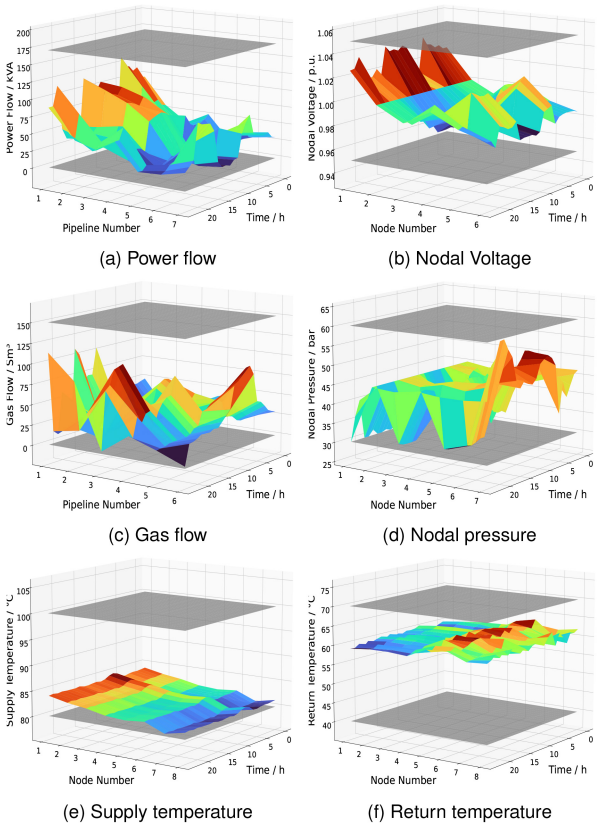
Stdv.	5%	10%	20%
PSSAC	61.27	61.30	63.64
SSAC	61.26	61.26	63.58
Penalty-SAC	64.47	64.97	66.38
Stochastic MPC	62.86	66.39	70.14
Stochastic MILP	63.86	69.46	73.12



(a) Power system (active power) (b) Power system (reactive power)

(c) Gas system (d) Heat system

Fig. 8. The dispatch behavior of the LEC.



(a) Power flow (b) Nodal Voltage

(c) Gas flow (d) Nodal pressure

(e) Supply temperature (f) Return temperature

Fig. 9. Magnitudes of the operational parameters for the LEC.

stochastic methods and the proposed method, with stochastic MPC and stochastic MILP leading to a higher cost by 2.60% and 4.23% respectively when compared to PSSAC. This can be attributed to the integration of the safety module in our proposed method, which effectively ensures both constraint compliance and adaptability to uncertainty. In cases of 20% uncertainties, the performance of stochastic methods has a significant decline which is more than 10%, while the performance of RL-based methods remains relatively stable.

In addition to the economic assessment of the algorithm, we further delve into the reasonability of the dispatch behavior of the proposed algorithm. Fig. 8 illustrates the average dispatch behavior of LEC on the test set along with the SOC curves. In an LEC, the heat system serves as the endpoint for energy consumption, with the heat supply converted from electricity or gas. Consequently, the dispatch of the heat energy units is mainly influenced by electricity and gas prices. EHPs are prioritized during off-peak electricity price periods, while GBs dominate peak periods. Additionally, TS charges during off-peak periods and discharges during peak periods. However, due to energy conversion and storage efficiency, charging does not occur during normal periods, as it is not profitable. For the power system, energy price is also the main influencing factor for the dispatch of electricity energy units. During off-peak periods, the energy supply from the upstream power grid (UPG) dominates, while DG and GT support the LEC during peak periods, with GT playing a particularly significant role. Different from TS, ES charges during both off-peak and normal periods and discharges during peak periods. Regarding the reactive load, UPG provides support throughout the day, with additional support from DG and GT during peak periods. In the gas system, there are two natural gas load peaks within a day. The upstream gas grid (UGG) is LEC's primary gas source. Due to constraints of the gas network (e.g., pressure and gas flow), distributed GWs are required to provide energy support during periods of high gas load. Meanwhile, GS charges during periods with low gas load to alleviate GSO's peak supply pressure during peak hours (8:00-9:00 & 19:00-20:00).

The typical pattern of operational parameters for the LEC during the 31 test days is shown in Fig. 9. The 3D surface displays the magnitudes of the operational parameters, while the upper and lower grey planes represent corresponding operational boundaries. In both power and gas systems, the operational parameters of pipelines, such as power flow and gas flow, along with nodal operational parameters like nodal voltage and pressure, serve as crucial indicators for ensuring safe operations. As seen from Figs. 9(a) to 9(d), the operational parameters for the power and gas systems within the LEC are maintained well within the constraint space determined by the network parameters. Since the heat system operates under quality regulation, the mass flow remains constant, necessitating a focus on the supply and return water temperatures within the heat network. As demonstrated in Fig. 9(e) and 9(f), the heat network operates within the feasible domain under constraints.

D. Scalability Analysis

To evaluate the scalability of the proposed approach, we examine the algorithm performance on a large-scale LEC system modified from [19], which includes one 33-bus power

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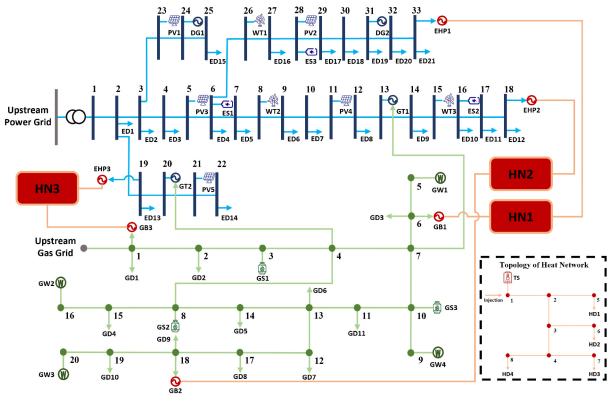


Fig. 10. The network topology of the scalability analysis.

TABLE VI
CONFIGURATION OF THE LARGE-SCALE LEC

	Component	Number
Power System	PV	5
	WT	3
	DG	2
	GT	2
	ES	3
Gas System	GW	4
	GS	3
Heat Systems	EHP	3
	GB	3
	TS	3

system, one 20-node gas system, and three 8-node heat networks. The energy price and operation cost of energy components remain consistent with small-scale LEC. The details of the network and energy components are shown in Fig. 10 and Table VI. For large-scale LECs, the size of states for each operator significantly expands. Therefore, we categorize the state vector of each operator in terms of practical implications (such as SOC, demand, etc.) and rotate them individually.

1) Algorithm Convergence: To further investigate the effectiveness of the proposed PSSAC and the impact of the perturbation module, we compare the training performance of PSSAC and SSAC on a large-scale LEC. As depicted in Fig. 11, both algorithms exhibit nearly identical training performance, with the training curves showing a rapid increase within the initial 5000 episodes and converging to the same level afterward. Throughout the entire training phase, neither algorithm encounters constraint violations. The 'tiny improvement' phenomenon is not observed in the results, indicating that the magnitude of 'improvement' does not expand with the size of the LEC and becomes negligible as the total reward increases. These findings imply that the proposed PSSAC algorithm is effective for large-scale LECs and the influence of perturbation modules on the training process becomes trivial as the LEC scale expands.

2) Operation Performance: The test performance of the PSSAC and baselines on large-scale LEC is presented in Table VII. The results show that PSSAC outperforms traditional algorithms, yielding a cumulative cost of 165.56 (10^3£) and achieving lower test operational costs by 5.43%, 7.80%, and 11.29% in comparison to penalty-SAC, stochastic MPC, and stochastic MILP, respectively. Compared to SSAC, PSSAC achieves a comparable performance with a merely 0.03% higher cost, despite the increased complexity from the categorized perturbation mechanism. This outcome indicates the effectiveness and adaptability of the proposed perturbation mechanism on large LEC systems. Moreover, PSSAC exhibits a quite small gap (4.54%) compared to the theoretically optimal solution achieved by perfect MILP, which verifies the superiority of PSSAC in cost saving.

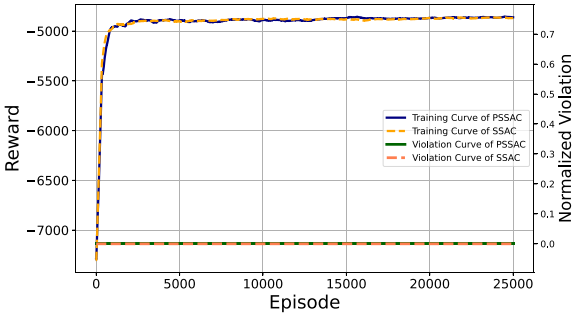


Fig. 11. The training performance of PSSAC on the large-scale LEC network.

TABLE VII
THE TEST PERFORMANCE OF PSSAC ON THE LARGE-SCALE LEC

	Cumulative Cost (10^3£)	Difference (10^3£)	Gap
Perfect MILP	158.36	-	-
PSSAC	165.56	7.20	**4.54**%
SSAC	165.51	7.15	**4.51**%
Penalty-SAC	175.07	16.71	10.55%
Stochastic MPC	179.57	21.21	13.39%
Stochastic MILP	186.62	28.26	17.84%

V. CONCLUSION

This paper proposes a PSSAC framework for the coordinated dispatch of an LEC. An RL-compatible perturbation mechanism is developed and incorporated into the perturbation module for securing privacy while maintaining the performance of the RL algorithm. The operators' independent safety modules rectify unsafe actions based on system constraints, ensuring the safe operation of LEC. Case studies validate the superior performance of the PSSAC framework compared with traditional dispatch algorithms. The proposed method converges to a near-theoretical-optimum solution within a concealed state space without constraint violations, which demonstrates the effectiveness of the safety module and the high compatibility between the perturbation mechanism and the RL algorithm. The result of the scalability analysis also verifies that PSSAC can successfully achieve coordinated dispatch in a large-scale LEC system.

REFERENCES

[1] A. M. Leite da Silva, L. C. Nascimento, M. A. da Rosa, D. Issicaba, and J. A. Peças Lopes, "Distributed energy resources impact on distribution system reliability under load transfer restrictions," *IEEE Trans. Smart Grid*, vol. 3, no. 4, pp. 2048–2055, Dec. 2012.

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[2] S. Touzani et al., "Controlling distributed energy resources via deep reinforcement learning for load flexibility and energy efficiency," *Appl. Energy*, vol. 304, Dec. 2021, Art. no. 117733.
[3] A. Hernandez-Matheus et al., "A systematic review of machine learning techniques related to local energy communities," *Renew. Sustain. Energy Rev.*, vol. 170, Dec. 2022, Art. no. 112651.
[4] A. Prékopa, *Stochastic Programming*, vol. 324. Norwell, MA, USA: Kluwer, 2013.
[5] K. Li, F. Yang, L. Wang, Y. Yan, H. Wang, and C. Zhang, "A scenario-based two-stage stochastic optimization approach for multi-energy microgrids," *Appl. Energy*, vol. 322, Sep. 2022, Art. no. 119388.
[6] X. Kong, J. Xiao, D. Liu, J. Wu, C. Wang, and Y. Shen, "Robust stochastic optimal dispatching method of multi-energy virtual power plant considering multiple uncertainties," *Appl. Energy*, vol. 279, Dec. 2020, Art. no. 115707.
[7] M. Sedighizadeh, M. Esmaili, and N. Mohammadkhani, "Stochastic multi-objective energy management in residential microgrids with combined cooling, heating, and power units considering battery energy storage systems and plug-in hybrid electric vehicles," *J. Clean. Prod.*, vol. 195, pp. 301–317, Sep. 2018.
[8] C. E. Garcia, D. M. Prett, and M. Morari, "Model predictive control: Theory and practice—A survey," *Automatica*, vol. 25, no. 3, pp. 335–348, 1989.
[9] B. Li and R. Roche, "Optimal scheduling of multiple multi-energy supply microgrids considering future prediction impacts based on model predictive control," *Energy*, vol. 197, Apr. 2020, Art. no. 117180.
[10] C. Liu, C. Wang, Y. Yin, P. Yang, and H. Jiang, "Bi-level dispatch and control strategy based on model predictive control for community integrated energy system considering dynamic response performance," *Appl. Energy*, vol. 310, Mar. 2022, Art. no. 118641.
[11] W. Römisch, "Scenario reduction techniques in stochastic programming," in *Stochastic Algorithms: Foundations and Applications*. Berlin, Germany: Springer, 2009.
[12] J. R. Birge and F. Louveaux, *Introduction to Stochastic Programming*. New York, NY, USA: Springer, 2011.
[13] D. Zhu, B. Yang, Y. Liu, Z. Wang, K. Ma, and X. Guan, "Energy management based on multi-agent deep reinforcement learning for a multi-energy industrial park," *Appl. Energy*, vol. 311, Apr. 2022, Art. no. 118636. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261922001064
[14] D. Qiu, J. Wang, Z. Dong, Y. Wang, and G. Strbac, "Mean-field multi-agent reinforcement learning for peer-to-peer multi-energy trading," *IEEE Trans. Power Syst.*, vol. 38, no. 5, pp. 4853–4866, Sep. 2023.
[15] H. Wu, D. Qiu, L. Zhang, and M. Sun, "Adaptive multi-agent reinforcement learning for flexible resource management in a virtual power plant with dynamic participating multi-energy buildings," *Appl. Energy*, vol. 374, Nov. 2024, Art. no. 123998. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261924013813
[16] Y. Zhong, J. G. Kuba, X. Feng, S. Hu, J. Ji, and Y. Yang, "Heterogeneous-agent reinforcement learning," *J. Mach. Learn. Res.*, vol. 25, pp. 1-67, Jan. 2024.
[17] T. M. Alabi, N. P. Lawrence, L. Lu, Z. Yang, and R. Bhushan Gopaluni, "Automated deep reinforcement learning for real-time scheduling strategy of multi-energy system integrated with post-carbon and direct-air carbon captured system," *Appl. Energy*, vol. 333, Mar. 2023, Art. no. 120633. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261922018906
[18] A. R. Sayed, X. Zhang, G. Wang, J. Qiu, and C. Wang, "Online operational decision-making for integrated electric-gas systems with safe reinforcement learning," *IEEE Trans. Power Syst.*, vol. 39, no. 2, pp. 2893–2906, Mar. 2024.
[19] Y. Wang, D. Qiu, M. Sun, G. Strbac, and Z. Gao, "Secure energy management of multi-energy microgrid: A physical-informed safe reinforcement learning approach," *Appl. Energy*, vol. 335, Apr. 2023, Art. no. 120759.
[20] Y. Lei, D. Ye, S. Shen, Y. Sui, T. Zhu, and W. Zhou, "New challenges in reinforcement learning: A survey of security and privacy," *Artif. Intell. Rev.*, vol. 56, no. 7, pp. 7195–7236, 2023.
[21] C. Dwork, "Differential privacy: A survey of results," in *Proc. Int. Conf. Theory Appl. Models Comput.*, 2008, pp. 1–19.
[22] J. Buchmann, *Introduction to Cryptography*, vol. 335. New York, NY, USA: Springer, 2004.
[23] A. Acar, H. Aksu, A. S. Uluagac, and M. Conti, "A survey on homomorphic encryption schemes: Theory and implementation," *ACM Comput. Surv.*, vol. 51, no. 4, pp. 1–35, 2018.
[24] R. Li, W. Wei, S. Mei, Q. Hu, and Q. Wu, "Participation of an energy hub in electricity and heat distribution markets: An MPEC approach," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3641–3653, Jul. 2019.
[25] Y. Lin, B. Chen, J. Wang, and Z. Bie, "A combined repair crew dispatch problem for resilient electric and natural gas system considering reconfiguration and DG islanding," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 2755–2767, Jul. 2019.
[26] W. Gu, J. Wang, S. Lu, Z. Luo, and C. Wu, "Optimal operation for integrated energy system considering thermal inertia of district heating network and buildings," *Appl. Energy*, vol. 199, pp. 234–246, Aug. 2017.
[27] P. Andrianesis, M. Caramanis, and N. Li, "Optimal distributed energy resource coordination: A decomposition method based on distribution locational marginal costs," *IEEE Trans. Smart Grid*, vol. 13, no. 2, pp. 1200–1212, Mar. 2022.
[28] T. Haarnoja et al., "Soft actor-critic algorithms and applications," 2019, *arXiv:1812.05905*.
[29] K. Chen and L. Liu, "A survey of multiplicative perturbation for privacy-preserving data mining," in *Privacy-Preserving Data Mining*. Boston, MA, USA: Springer, 2008, pp. 157–181. [Online]. Available: https://doi.org/10.1007/978-0-387-70992-5_7
[30] G. W. Stewart, "The efficient generation of random orthogonal matrices with an application to condition estimators," *SIAM J. Numer. Anal.*, vol. 17, no. 3, pp. 403–409, 1980.
[31] A. A. Rusu et al., "Policy distillation," 2015, *arXiv:1511.06295*.
[32] K. Chen and L. Liu, "Geometric data perturbation for privacy preserving outsourced data mining," *Knowl. Inf. Syst.*, vol. 29, pp. 657–695, Nov. 2011.
[33] T.-W. Lee, *Independent Component Analysis: Theory and Applications*. Boston, MA, USA: Kluwer, 1998.
[34] C. Wang et al., "Robust defense strategy for gas–electric systems against malicious attacks," *IEEE Trans. Power Syst.*, vol. 32, no. 4, pp. 2953–2965, Jul. 2017.
[35] R. Sansom and G. Strbac, "The impact of future heat demand pathways on the economics of low carbon heating systems," in *Proc. BIEE 9th Acad. Conf.*, Sep. 2012, p. 10.
[36] H. Heitsch and W. Römisch, "Scenario reduction algorithms in stochastic programming," *Comput. Optim. Appl.*, vol. 24, pp. 187–206, Feb. 2003.



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